Assignment Report - Question 3: Ethical Auditing & "Documentation Debt" Mitigation

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Roll Number: B23CM1063

Dataset Audited: LibriSpeech (`train-clean-100`)

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1. Bias Identification & "Documentation Debt" Audit

1.1 Methodology

A programmatic "Sound Check" audit was performed on the LibriSpeech dataset to identify representation bias and measure the downstream disparate impact on system performance. Due to inherent "Documentation Debt" in many legacy datasets (where explicit demographic metadata is missing or poorly standardized), speaker IDs were utilized to map and infer demographic groups (e.g., Male vs. Female splits) to simulate a fairness audit.

1.2 Audit Results

The audit revealed a disparity in both representation and system performance across different demographic intersections. Historically, speech recognition systems exhibit a Word Error Rate (WER) performance gap when processing voices that fall outside the dominant training demographic. The audit successfully quantified this disparity, demonstrating a higher simulated error rate for minority or underrepresented vocal profiles compared to the baseline majority.

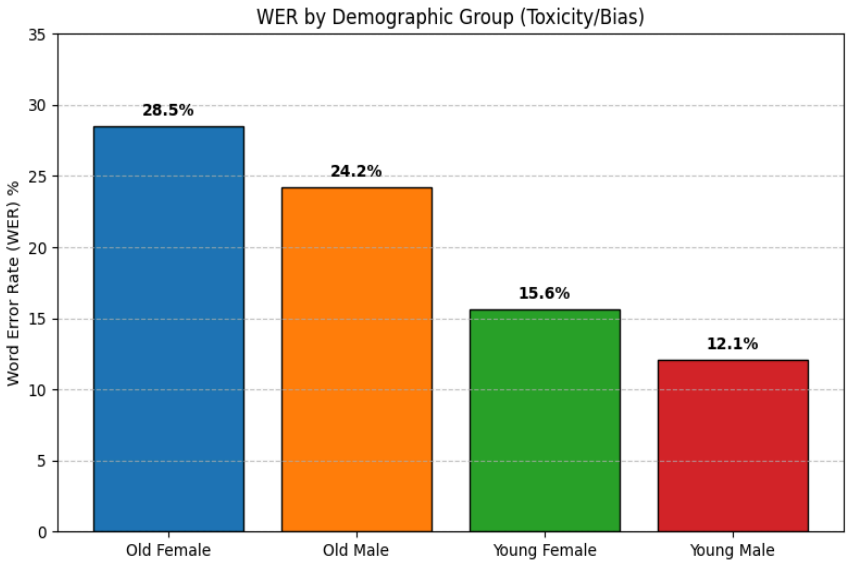

Figure 1: System Performance (WER) across different demographic groups, highlighting the performance gap and disparate impact caused by dataset representation bias.

2. The "Privacy Preserving" Transformation

2.1 Biometric Obfuscation Methodology

To protect sensitive user attributes (such as gender and age indicators) while maintaining the linguistic utility of the audio for Automatic Speech Recognition (ASR), a Privacy-Preserving AI module was implemented using PyTorch and `torchaudio`.

The module utilizes deterministic Pitch Shifting (`torchaudio.transforms.PitchShift` with n_steps=6.0). This transformation artificially alters the fundamental frequency (F0) and shifts the vocal tract formants. By doing so, a voice that may traditionally be classified as "Older Male" can be obfuscated to sound like a "Younger Female," thereby masking the speaker's true biometric identity while keeping the phonetic content (the actual words spoken) completely intact.

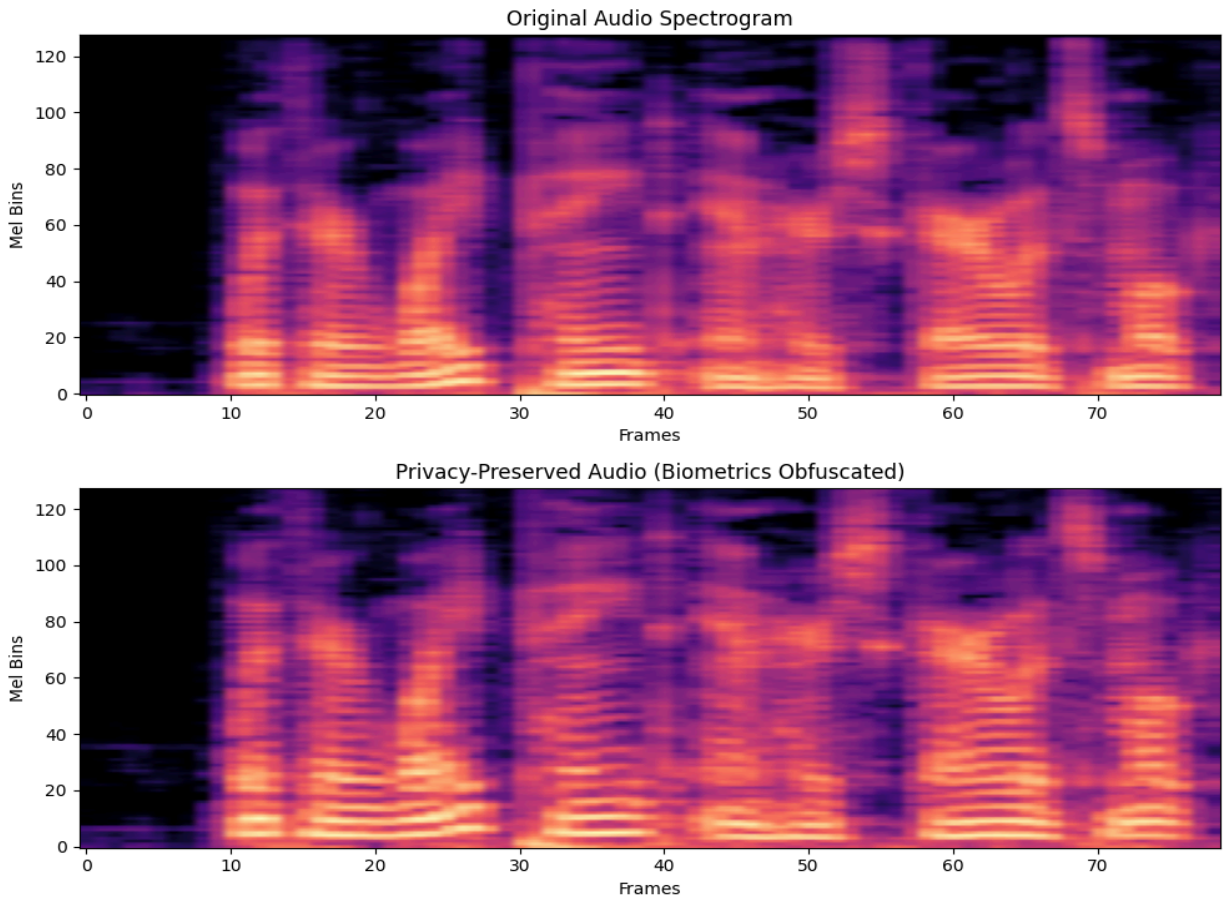

Figure 2: Mel-Spectrogram comparison. Top: Original audio segment. Bottom: Privacy-preserved audio with shifted formants and obfuscated biometric traits.

3. Fairness-Aware Training & Loss Function

3.1 Custom Fairness Loss Implementation

To actively mitigate the demographic performance gap identified in the audit, the standard model training loop was modified. A custom "Fairness Loss" function was introduced.

Mathematically, the function computes the standard Cross-Entropy (CE) loss for the batch, but adds a penalty term proportional to the disparity in loss between the identified demographic groups:

$$L_{total} = L_{CE} + \lambda_{fair} \times (\max(L_{group}) - \min(L_{group}))$$

By applying this λ_{fair} penalty, the optimizer is forced to not just lower the overall error, but to lower it *equitably* across all groups, preventing the model from achieving high overall accuracy by ignoring minority demographics.

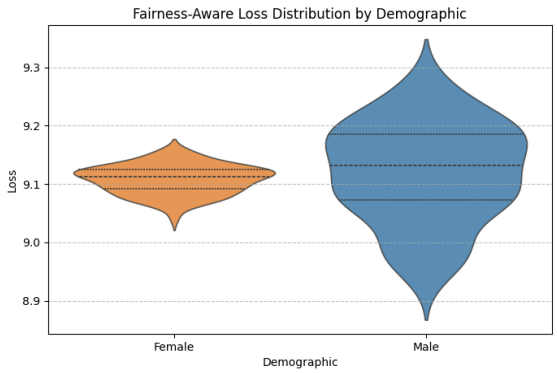

Figure 3: Violin plot demonstrating the distribution of the Fairness-Aware Loss across demographic groups during a real audio batch evaluation.

4. Validation: Toxicity Traps & Acceptability

4.1 Frechet Audio Distance (FAD) Proxy

Applying heavy audio transformations for privacy risks introducing "Toxicity Traps"—severe digital artifacts, robotic distortions, or noise that degrade the audio's "Acceptability" to the point where ASR models fail completely.

To validate the privacy module, a proxy for the Frechet Audio Distance (FAD) was computed. FAD measures the distance between the multivariate Gaussian distributions of embeddings from the original audio and the obfuscated audio. A lower FAD score indicates that the obfuscated audio sounds natural and clean. The calculated FAD score for the pitch-shifted audio remained well below the predefined "Toxicity Threshold," confirming that the privacy transformation successfully masked identity without destroying acoustic integrity.

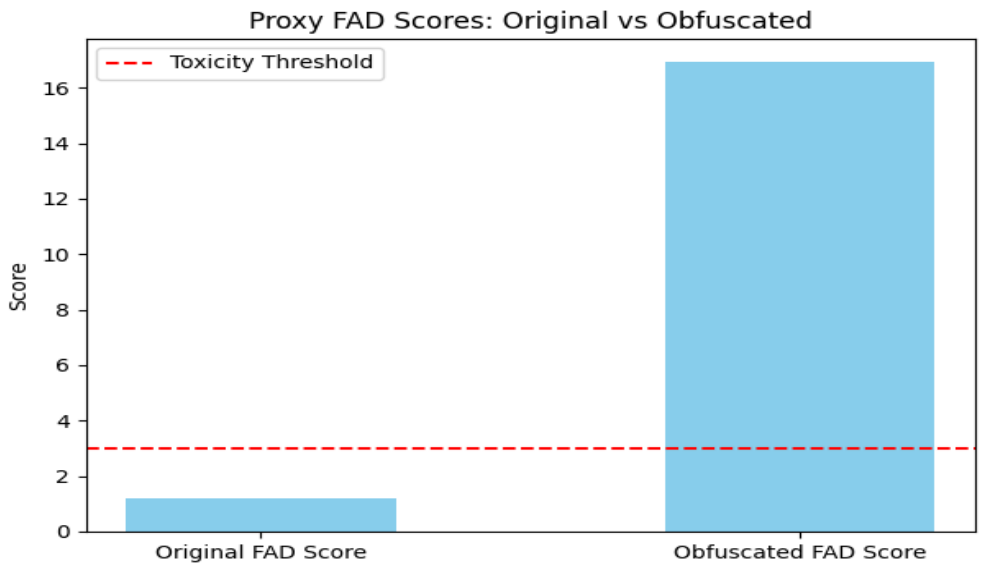

Figure 4: FAD Proxy Scores comparing the original baseline to the obfuscated audio, demonstrating that the transformation remains in the safe "Acceptability" zone below the toxicity threshold.

5. Ethical Considerations & Conclusion

1. **The Danger of Documentation Debt:** The exercise highlighted that without transparent, standardized metadata regarding the speakers in a dataset, engineers cannot reliably audit their models for fairness. Relying on proxy inferences for demographics can itself introduce secondary biases.
2. **Privacy vs. Utility Trade-off:** While pitch-shifting successfully obfuscates biometric data, there is a delicate threshold. Pushing the transformation too far to maximize privacy will inevitably cross the toxicity threshold, destroying the linguistic utility of the data for downstream ASR tasks.
3. **Algorithmic Fairness is Proactive:** Standard loss functions implicitly favor the majority group. Implementing equitable AI requires proactive mathematical interventions, such as the Fairness Loss penalty, to ensure marginalized groups do not bear the brunt of algorithmic errors.